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L04: Reflection Journal – Text Representation
ITAI 2373 – NLP
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Reflective Journal: Text Representation

Introduction

In this lab, I explored multiple text representation techniques including Bag of Words (BOW), TF-IDF, n-grams, and word embeddings. The purpose of this reflection is to examine how my understanding of text representation evolved, how the technical implementation affected my learning, and what insights I gained about both performance and ethical considerations.

Description of Experience

Throughout the five parts of this lab, I progressed from simple frequency-based representations (BOW) to more advanced semantic representations (TF-IDF and word embeddings). I implemented feature extraction methods, compared classification performance, and analyzed dimensionality differences between sparse and dense representations.
The most impactful part of the experience was observing how representation choices directly influence model behavior and interpretability. Initially, text seemed like a collection of words to count. By the end, it became clear that representation determines how meaning is encoded mathematically.

Analysis and Interpretatio

What surprised me most was how embeddings capture relationships mathematically. The example:

- king − man + woman ≈ queen

demonstrated that vector space can encode conceptual relationships. That insight changed how I view NLP models. They are not just counting words; they are mapping relationships in high-dimensional space.

I also learned that performance similarities (such as similar accuracy between BOW and TF-IDF in this lab) do not necessarily mean the representations are equally powerful. TF-IDF provides more meaningful weighting by downscaling common words and emphasizing distinctive ones.

Connection to Theory

This lab reinforced the distributional hypothesis and demonstrated the difference between sparse and dense vector spaces. Sparse representations preserve interpretability but lack deep semantic encoding. Dense embeddings sacrifice direct interpretability in exchange for richer semantic structure.

This aligns with the broader NLP shift from symbolic methods to representation learning.

Critical Evaluation

What worked well:

- Clear comparison between sparse and dense models
- Visualization of feature counts and accuracy
- Ethical discussion around bias in embeddings

What was challenging:

- Debugging classification reports
- Understanding how embeddings mathematically encode meaning
- Managing Markdown formatting issues in the notebook

If I were to improve this process, I would:

- Write TF-IDF from scratch to fully internalize its mechanics
- Experiment with larger datasets to observe performance divergence
- Visualize embedding space using dimensionality reduction (e.g., PCA or t-SNE)

Discussion of Improvements and Learning

Personal Growth

This lab shifted my understanding from "text = words" to "text = structured numerical representation of meaning." That is a major conceptual upgrade.

I also strengthened my debugging resilience. Technical friction (especially formatting and classification report errors) forced me to think carefully about labels, outputs, and structure.

Skills Developed

- Implementing BOW and TF-IDF in practice
- Comparing sparse vs dense representations
- Interpreting classification metrics
- Recognizing ethical implications in NLP systems
- Debugging model output inconsistencies

Future Application

Going forward, I want to explore:

- Building TF-IDF manually to deeply understand weighting
- Transformer-based embeddings (BERT, contextual embeddings)
- Bias measurement techniques in embedding spaces
- Semantic similarity and vector clustering
- Real-world deployment concerns in AI systems

These areas would help me bridge academic understanding with real-world NLP systems.

Conclusion

Over the course of this lab, my understanding of text representation evolved from surface-level word counting to conceptual appreciation of vector space semantics. The most significant insight was realizing that meaning can be encoded mathematically and manipulated through vector arithmetic.

This lab strengthened both my technical skills and my conceptual understanding of NLP systems. It also reinforced the importance of ethical awareness when deploying models trained on real-world language data.